IT4010 Research Project – 2026 Jul

Topic Assessment Form

Project ID: **J26-IT-343**

1. **Topic (12 words max)**

AI-Driven Quality Assurance and Blockchain-Based Traceability Framework for Sri Lankan Fruit Export Industry

2. **Research group the project belongs to**

CoEAI - Centre of Excellence for AI

3. **Specialization of the project belongs to**

Information Technology (IT)

4. **If a continuation of a previous project:**

Project ID	N/A
Year	N/A

5. **Brief description of the research problem including references (300 words max) – references not included in word count.**

The Sri Lankan fruit export industry faces significant challenges in maintaining product quality, predicting shelf life, and ensuring transparent traceability throughout the supply chain. Quality inspection is primarily performed through manual visual assessment by human inspectors, which is subjective, time-consuming, and susceptible to inconsistencies. Consequently, defective, diseased, or low-quality fruits may be included in export shipments, resulting in financial losses, reduced customer satisfaction, and rejection by international markets [1].

Recent studies have demonstrated the effectiveness of deep learning and computer vision techniques for fruit classification and defect detection. Convolutional Neural Networks (CNNs) have achieved high accuracy in identifying fruit diseases, bruises, and surface defects from images [2][3]. However, most existing solutions focus solely on defect detection and do not provide comprehensive quality assessment combined with shelf-life prediction. Since fruit deterioration is influenced by multiple factors such as maturity level, storage conditions, temperature, and humidity, accurate shelf-life estimation remains a challenging problem [4].

Furthermore, traceability within agricultural export supply chains is often fragmented due to reliance on centralized databases and manual record-keeping processes. These approaches can lead to data manipulation, lack of transparency, and difficulties in tracking the origin and handling history of exported products [5]. Blockchain technology has emerged as a promising solution for ensuring data integrity and end-to-end traceability in agri-food supply chains; however, existing blockchain-based systems rarely integrate AI-powered quality assessment and predictive analytics into a unified framework [6].

Therefore, there is a need for an integrated platform that combines AI-based fruit quality assessment, machine learning-driven shelf-life prediction, and blockchain-enabled traceability. Such a system can improve export quality standards, reduce post-harvest losses, enhance consumer trust, and strengthen the competitiveness of Sri Lankan fruit exports in international markets.

References

[1] Food and Agriculture Organization (FAO), The State of Food and Agriculture, Rome, Italy, 2023.

[2] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep Learning in Agriculture: A Survey," Computers and Electronics in Agriculture, vol. 147, pp. 70–90, 2018.

[3] K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," Proceedings of CVPR, pp. 770–778, 2016.

[4] M. A. Rahman et al., "Machine Learning Approaches for Fruit Shelf-Life Prediction: A Review," Postharvest Biology and Technology, 2023.

[5] S. Tian, "An Agri-food Supply Chain Traceability System for China Based on RFID and Blockchain Technology," International Journal of Distributed Sensor Networks, vol. 12, no. 11, 2016.

[6] M. Caro, M. Ali, M. Vecchio and R. Giaffreda, "Blockchain-based Traceability in Agri-Food Supply Chain Management," Future Internet, vol. 10, no. 4, 2018.

6. Brief description of Existing Research and Systems (300 words max)

Numerous research studies have explored the use of Artificial Intelligence (AI) and Computer Vision techniques for fruit quality assessment and disease detection. Deep learning models such as Convolutional Neural Networks (CNNs), ResNet, and EfficientNet have demonstrated high accuracy in identifying fruit defects, diseases, bruises, and ripeness levels from image datasets [1][2]. These systems help reduce human intervention and improve inspection efficiency compared to traditional manual methods.

Several commercial solutions and smart agriculture platforms also provide quality monitoring and supply chain management capabilities. Companies such as IBM Food Trust and other blockchain-enabled agricultural platforms utilize blockchain technology to improve food traceability and transparency throughout the supply chain [3]. These systems allow stakeholders to track product origins and movement records, enhancing consumer trust and regulatory compliance.

Researchers have additionally investigated machine learning techniques for predicting fruit shelf life using environmental factors such as temperature, humidity, storage conditions, and fruit maturity levels [4]. These approaches help reduce post-harvest losses and improve inventory management. However, most studies focus on shelf-life prediction as an independent component without integrating quality assessment and traceability mechanisms.

Despite these advancements, significant limitations remain. Existing AI-based systems mainly focus on defect detection or disease classification and rarely provide comprehensive quality scoring. Blockchain-based traceability solutions generally lack integration with AI-powered inspection results and predictive analytics. Furthermore, many existing systems are designed for large-scale agricultural industries and are not specifically tailored to the requirements of Sri Lanka's fruit export sector.

Therefore, there is a research gap for a unified framework that combines AI-driven fruit quality assessment, machine learning-based shelf-life prediction, and blockchain-enabled traceability. Addressing this gap can improve export quality assurance, reduce food waste, and provide transparent supply chain management for Sri Lankan fruit exporters.

References

[1] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep Learning in Agriculture: A Survey," Computers and Electronics in Agriculture, 2018.

[2] K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, 2016.

[3] M. Caro, M. Ali, M. Vecchio and R. Giaffreda, "Blockchain-based Traceability in Agri-Food Supply Chain Management," Future Internet, 2018.

[4] M. A. Rahman et al., "Machine Learning Approaches for Fruit Shelf-Life Prediction: A Review," Postharvest Biology and Technology, 2023.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed research presents an AI/ML-based Smart Fruit Quality Inspection, Shelf-Life Prediction, and Traceability System that aims at enhancing the efficiency and effectiveness of the fruit export business in Sri Lanka. Currently, there is no technological approach towards inspecting the quality of fruits that can be exported; most of these assessments depend on manual procedures, which might involve inconsistencies in their process.

In order to solve this problem, the proposed technology makes use of Computer Vision, Deep Learning, Machine Learning, Blockchain Technology, and Mobile Computing in its implementation. With the help of the technology, one is able to upload images of fruits using a mobile application, whereupon the system provides assessment with regard to various aspects such as quality, defects, variety, shelf-life, etc.

The proposed solution is comprised of four distinct intelligent modules. The first one detects defects and diseases within the fruit and evaluates the level of severity, as well as assigns fruit quality grade. The second module detects fruit varieties and analyzes its export readiness based on specific variety standards. The third module makes predictions about the fruit remaining shelf life based on visual features taken from pictures. The fourth module offers blockchain-based traceability, supply chain analytics, and export readiness support via QR code tracking.

This work solves the identified problem by substituting subjective manual inspection procedures with AI-based analysis, providing reliable quality assessment, defect and disease detection early on, shelf-life prediction without the need for extra hardware sensors, and blockchain-based traceability of the entire supply chain process. This solution is purely software-based.

System Perspective (Input – Process – Output)

Input

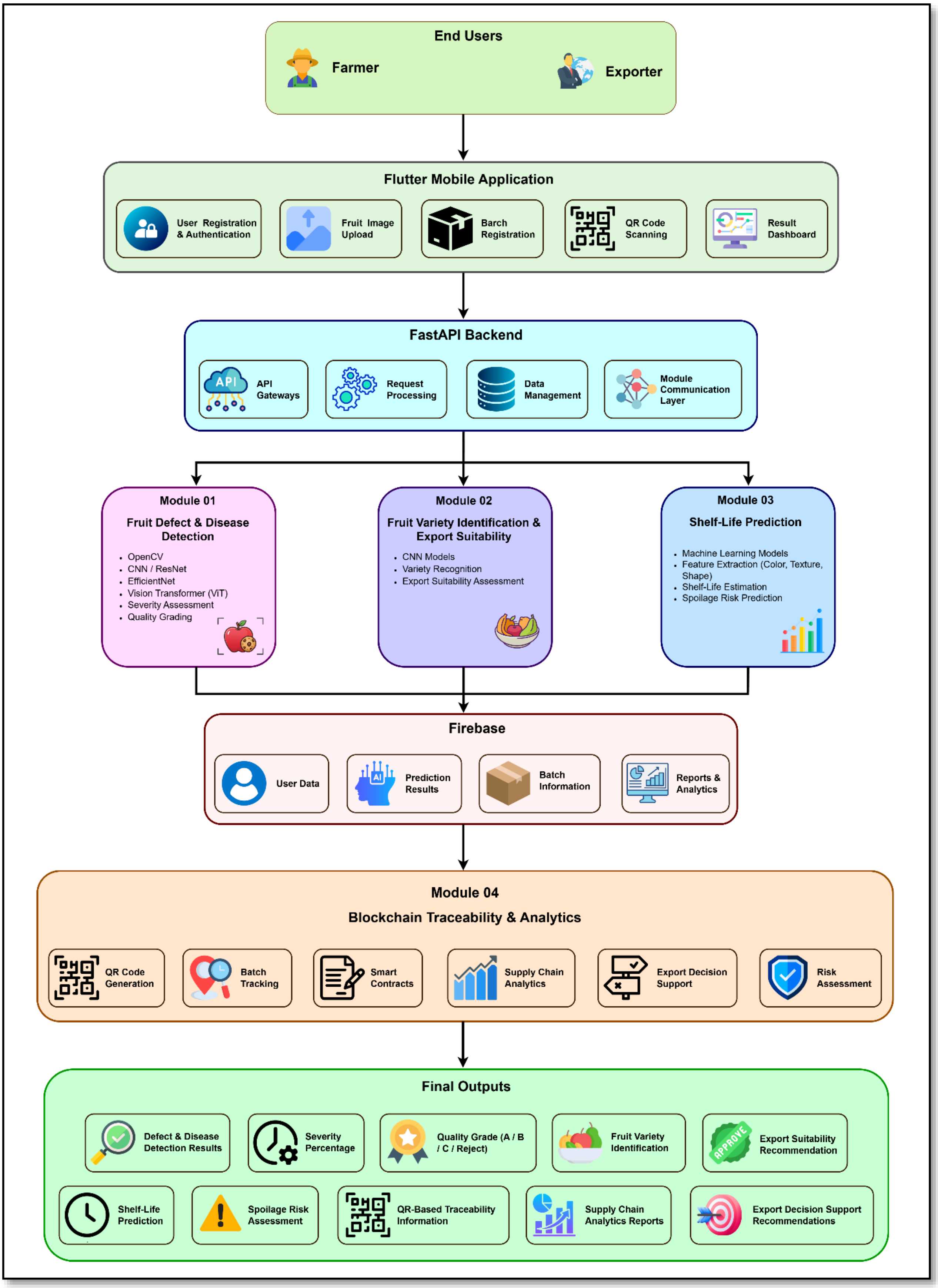
- Fruit images uploaded via mobile application
- Batch registration information
- Supply chain and export-related records

Process

- Image preprocessing using OpenCV
- Defect and disease detection using Deep Learning models
- Severity assessment and quality grading
- Fruit variety identification
- Shelf-life prediction using Machine Learning algorithms
- Blockchain-based traceability and analytics
- Export suitability evaluation and recommendation generation

Output

- Detected defect and disease categories
- Severity percentage and risk assessment
- Quality grade (Grade A, Grade B, Grade C, Reject)
- Fruit variety identification results
- Predicted remaining shelf life
- Export suitability recommendations
- QR-based traceability reports and supply chain insights
- Comprehensive quality assessment reports



Link: High-Resolution Conceptual Diagram

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Domain Expertise

• Expertise in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision is required to develop fruit quality assessment, defect detection, disease identification, and shelf-life prediction models.

• Knowledge of Blockchain technology and Smart Contracts is necessary to implement secure and transparent traceability mechanisms across the fruit export supply chain.

• Agricultural and post-harvest management knowledge is required to understand fruit grading standards, quality indicators, disease characteristics, ripeness levels, and factors affecting shelf life.

• Knowledge of software engineering, cloud computing, database management, and web/mobile application development is needed to build an integrated export quality assurance platform.

Knowledge

• Understanding of deep learning frameworks such as TensorFlow, PyTorch, and OpenCV for image processing and defect detection.

• Knowledge of predictive analytics and machine learning algorithms for shelf-life estimation based on environmental and quality-related parameters.

• Familiarity with blockchain platforms such as Ethereum or Hyperledger Fabric for developing immutable traceability records.

• Understanding of REST APIs, cloud deployment, database systems, and system integration technologies.

• Knowledge of export quality regulations, food safety standards, and traceability requirements applicable to international fruit markets.

Data Requirements

• Fruit Image Dataset: Images of fresh, defective, diseased, bruised, and export-quality fruits collected from farms, exporters, and publicly available agricultural datasets.

• Environmental Data: Temperature, humidity, storage conditions, transportation details, and other factors affecting fruit quality and shelf life.

• Historical Export Data: Quality inspection reports, rejection records, and export performance data for model training and validation.

• Supply Chain Data: Farmer information, harvesting records, packaging details, logistics data, and export documentation required for blockchain-based traceability.

Availability of Expertise and Data

The required technical knowledge can be acquired through academic literature, online courses, research publications, and guidance from project supervisors. Agricultural expertise can be obtained through consultations with farmers, exporters, agricultural officers, and industry professionals. Data will be collected through field visits, collaborations with fruit exporters, publicly available agricultural datasets, and manually gathered supply chain records. Since the project primarily uses agricultural and operational data without collecting sensitive personal information, ethical risks are minimal; however, proper consent and data privacy measures will be followed when collecting industry-related information.

9. Objectives and Novelty

Main Objective:
Develop an AI-powered system for automated fruit quality assessment, shelf-life prediction, export suitability evaluation, and supply chain traceability in the Sri Lankan export industry.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Kulathunga R.K.M.D.V - IT23288430	AI-Based Fruit Defect and Disease Detection with Severity Assessment and Post-Harvest Quality Grading using Computer Vision	**1.1** Collect and annotate a comprehensive fruit image dataset containing defect types, disease categories, and severity levels. **1.2** Apply image preprocessing techniques such as background removal, image enhancement, segmentation, and noise reduction using OpenCV. **1.3** Develop and train Deep Learning models (CNN, ResNet, EfficientNet, ViT) to detect fruit defects and classify diseases with severity assessment. **1.4** Generate automated post-harvest quality grades (Grade A, Grade B, Grade C, Reject) based on detected defects, disease severity, and visual quality indicators. **1.5** Produce comprehensive quality assessment reports including defect maps, severity scores, export recommendations, and quality degradation analysis.	Multi-task AI model capable of simultaneously detecting fruit defects, identifying diseases, and estimating severity levels in a single prediction pipeline. Automated post-harvest quality grading aligned with Sri Lankan export standards and international market requirements. Explainable AI integration using Grad-CAM and LIME to visualize decision-making regions of defect detection models. Temporal quality degradation tracking through sequential image analysis for monitoring defect progression over storage periods.

Budara H M K IT23179158	AI-Based Shelf-Life Prediction using Multimodal Data	**2.1** Collect and build a dedicated dataset by capturing fruit images at regular storage intervals (e.g., daily) alongside corresponding actual shelf-life records, until visible spoilage occurs.	Most current shelf-life prediction systems rely on physical sensors or IoT devices to track temperature, humidity, or gas levels in storage. This component takes a different approach — predicting shelf life using only visual features from fruit images, without needing any sensor hardware. Because it depends purely on appearance-based data, the system becomes much simpler and cheaper to deploy in real-world settings, especially for small-scale farmers or vendors who can't afford sensor-based monitoring setups. To support this, a custom dataset was built by photographing fruits at regular time intervals and recording their actual spoilage timelines. Several machine learning algorithms were then trained and compared on this dataset to find the model that best predicts shelf life from visual cues alone. This makes the approach clearly different from existing multimodal or sensor-fusion systems, since it proves that shelf-life forecasting can be done effectively using image data by itself.
		2.2 Apply image preprocessing and visual feature extraction using OpenCV, including color histograms, texture patterns (GLCM, LBP), surface characteristics, and morphological properties to represent fruit appearance numerically.	
		2.3 Develop and train Machine Learning models (Random Forest, XGBoost, Support Vector Regression, Neural Networks) using Scikit-learn and TensorFlow to predict remaining shelf life (in days), spoilage risk level, and freshness duration based solely on extracted visual features.	
		2.4 Apply data augmentation and transfer learning techniques to overcome limited dataset size and improve model generalization across different fruit types and conditions.	
		2.5 Compare and evaluate all trained models using metrics such as RMSE, MAE, and R^2 score to identify the best-performing algorithm, and generate shelf-life prediction outputs including estimated remaining days, spoilage risk classification, and freshness reports.	

<table>
<tr>
<td rowspan="5">K D K Dilnuka
IT23143340</td>
<td rowspan="5">AI-Based Fruit Variety Identification and Classification for Export Suitability using Computer Vision and Deep Learning</td>
<td>3.1 Gather a large collection of fruit photos showing different varieties (like different types of bananas, mangoes, and avocados). Make sure the photos are taken in various lighting conditions, angles, and ripeness stages so the system learns to recognize them in real-world situations.</td>
<td rowspan="5">Identifying very similar-looking fruit types that belong to the same species - this is important because mixing up varieties can reduce market value and make buyers lose trust in the product.

Going beyond just naming the fruit - the system also tells whether that specific variety is good for export, which helps exporters make better business decisions.

Working together with the other parts of the system - variety information is combined with quality scores and traceability data, giving a complete picture of each fruit batch from farm to export.</td>
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<tr>
<td>3.2 Clean up the images and pull out important visual details using OpenCV. This includes removing backgrounds, adjusting colors, and extracting features like shape, color patterns, texture, and size so the AI can better tell different varieties apart.</td>
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<tr>
<td>3.3 Build and train AI models (using Convolutional Neural Networks) to recognize and classify different fruit varieties. For example, teach the system to tell the difference between Cavendish, Ambul, Kolikuttu, and Seeni bananas, or between different types of mangoes and avocados that are commonly exported.</td>
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<td>3.4 Check if each identified variety meets export quality standards and market requirements. Connect this variety information with the system's export decision-making tools to help exporters choose the right fruits for different markets.</td>
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<td>3.5 Generate clear results showing the fruit type, variety name, how confident the AI is about its identification, and whether that variety is suitable for export or not.</td>
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Shiromani N R A IT23160552	Blockchain-Based Traceability, Supply Chain Analytics, Export Decision Support, and Mobile App Integration	4.1 Design and implement a unique batch identification framework, QR code generation mechanism, and Hyperledger Fabric smart contracts to securely record and track fruit lifecycle data across the supply chain. 4.2 Integrate Firebase with the blockchain network to manage user authentication, cloud data storage, real-time synchronization, and QR-based batch verification at each supply chain checkpoint. 4.3 Build analytical modules to monitor batch performance, transportation history, spoilage trends, traceability compliance, and generate comprehensive reports for stakeholders. 4.4 Analyze historical export and logistics data to create recommendation algorithms that provide export readiness assessments, shipping route optimization, market suitability recommendations, and risk alerts. 4.5 Develop a mobile application that serves as a unified platform for QR scanning, blockchain traceability, analytics visualization, export management, and decision-support functionalities.	Fruit traceability systems that exist now mostly use QR codes or databases to track things. These systems do not give us a picture and can be easily tampered with. This study suggests using a blockchain system to track fruit batches from start to finish. The new thing about this system is that it uses Hyperledger Fabric and Firebase together. This means we get records that cannot be changed and data that is always up to date. Unlike systems this one tracks the fruit and also looks at the whole supply chain. It tells us things like how much fruit goes bad how well transport works and what is happening with each batch of fruit. This study also comes up with a way to help people decide when to export fruit. It looks at what happened in the past. Gives advice on when to ship, which route to take and what problems might come up. This helps people make decisions about exporting fruit. Another important thing is that this system puts everything together in

			one app. This means farmers, distributors, exporters and buyers can all use the system. The blockchain system, analytics and export management all work together. Can be accessed from a mobile phone. This is something for the fruit supply chain, especially, in Sri Lanka where they export a lot of fruit.

10. This section is to be completed by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Data required for the project is available.

Yes	✓	No	

f) Any other comments:

Recommended to continue the Research Project.

	Title	First Name	Last Name	Signature
Supervisor*	Mrs.	Lokesha	Weerasinghe	[signature]
Co-Supervisor*	Ms	Lashika	Chamini	[signature]
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

*- At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.